

Abdelhak BOURDIM

Senior Embedded Software Engineer

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Lorraine / Greater Region — Cross-border commuter

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[COMPANY NAME]

[Department]

[Address]

[Postal Code, Luxembourg]

[City], [Date]

Re: Application for the position of [JOB TITLE]

Dear Hiring Manager,

I am writing to express my keen interest in the [JOB TITLE] position at [COMPANY]. With **over 20 years of experience** in embedded software development, I am confident that my skills and background align closely with the requirements of this role. Based in **Lorraine**, I am available as a **cross-border commuter (frontalier)** and ready to work in Luxembourg immediately.

I specialize in **firmware, middleware and application layer development** — bare metal, RTOS-based (FreeRTOS, VxWorks) or Embedded Linux — building drivers and communication stacks (CAN, BLE, USB, TCP/IP) on ARM Cortex, STM32, Microchip and NXP architectures, within environments subject to stringent quality and certification requirements.

My career spans demanding industries that map directly onto Luxembourg's economic landscape: **aerospace and satellites** at Safran and Thales (DO-178C), skills transferable to projects at [Sector: e.g. SES / LuxSpace]; **medical devices** at GE Healthcare and Sorin (IEC 62304); **automotive and sensors** at E2-CAD (ISO 15118, V2G), relevant for companies such as IEE; **IoT cybersecurity** at Enedis (Linky smart meter audits and pentesting), aligned with INCERT and the telecom sector (POST Luxembourg); and **payment terminals** at Cartadis, where 8 years gave me a solid foundation in full-stack embedded development, from drivers to DES/RC4 encryption stacks. My code complies with MISRA C standards and my deliverables meet V-model traceability requirements.

What attracts me to [COMPANY] is [reason: e.g. your leadership in satellite communications / your innovation in automotive sensors / the critical nature of your industrial projects]. I am convinced that my experience in [relevant domain: e.g. DO-178 certified systems / embedded medical devices / secure IoT / EV charging] would bring significant value to your team. As a **trilingual professional (French, English, Arabic)**, I thrive in Luxembourg's multicultural and multilingual work environment.

Alongside my professional career, I founded **Workshop-DIY**, an educational robotics association where I lead workshops for young people on Arduino, ESP32, AI and 3D printing, reflecting my passion for knowledge sharing and innovation.

Available immediately, I would welcome the opportunity to discuss how my experience and expertise could contribute to your projects in greater detail.

Thank you for considering my application. I look forward to hearing from you.

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